Graphical Abstract

OASIS: Feedback-Driven Marginal Statistics Correction for Cost-Based Query Optimizers

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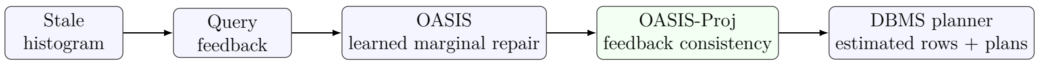

Highlights

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- OASIS repairs stale single-column marginal histograms without table rescans.
- PostgreSQL-style statistics are mapped to a canonical quantile view.
- Multi-column composition requires feedback-consistent marginals.
- Deployment safety is evaluated with traces, feedback noise, and planner checks.

OASIS: Feedback-Driven Marginal Statistics Correction for Cost-Based Query Optimizers

Author Names Redacted for Review[a]

[a] *Affiliation Redacted, , ,*

Abstract

Cost-based optimizers depend on column-level statistics that are refreshed by periodic `ANALYZE` passes. Between refreshes, continuous inserts, deletes, and updates create a staleness gap: the optimizer keeps planning with obsolete histograms even though query execution exposes predicate-level feedback. This paper presents OASIS, a feedback-driven layer that repairs stale single-column marginal histograms from recent query feedback without rescanning base tables or modifying the optimizer. OASIS maps engine-native statistics to a canonical quantile/CDF view, encodes the stale histogram and feedback as a compact tensor, and predicts corrected quantile boundaries with a lightweight pooled MLP.

The evaluation is optimizer-facing rather than runtime-facing. Across synthetic compound drift, OASIS reduces single-column selectivity Q-error by 51.3%, 57.9%, and 60.6% at drift intensities $q = 10$, 20, and 30, respectively. The results also identify an important deployment boundary: learned marginal repair improves one-dimensional statistics, but downstream multi-column composition is much more stable when the corrected marginals respect current feedback constraints. Plain OASIS has limited two-column composition gains, whereas OASIS-Proj calibrates the learned marginal before downstream use and recovers 31.8% and 23.1% joint Q-error improvements under independence and a Gaussian copula. In benchmark-inspired DML traces, aggregate Q-error falls from 1.295 for stale statistics to 1.076 with OASIS-Proj and 1.072 with residual-gated Hybrid. Across 12 PostgreSQL planner-only configurations, OASIS-Proj reduces aggregate row-estimation Q-error from 27.748 to 2.360 and matches the fresh-statistics plan shape on 96.1% of queries, without measuring query runtime.

Keywords: Statistics correction, Cardinality estimation, Histogram calibration, Cost-based optimizer, Query feedback, Online learning

1. Introduction

Cost-based optimizers (CBOs) choose query plans from estimates of predicate selectivity, intermediate cardinality, and operator cost. In many analytical database systems, these estimates rely on column-level statistics such as histograms, most-common-value lists, and density summaries. The statistics are periodically refreshed through `ANALYZE`-style procedures that read a full or sampled view of the table. This maintenance strategy is simple and robust, but it leaves a persistent staleness gap: tables continue to receive inserts, deletes, and updates while the optimizer keeps planning with the last statistics snapshot. As the gap widens, stale histograms induce systematic selectivity errors that can propagate through the optimizer's cost model.

Existing remedies operate at different layers of the optimization stack. Frequent `ANALYZE` reduces staleness but increases scan overhead. Learned cardinality estimators predict point cardinalities for particular queries and often require raw data, query workloads, or table-specific training. Plan-level feedback systems adjust plan choices or hints after observing execution behavior, but they do not repair the underlying statistics objects consumed by the optimizer. This paper focuses on a complementary layer: *statistics-level correction*. Rather than replacing the optimizer or learning complete query cardinalities, we update the column statistics supplied to the existing CBO.

The key observation is that a DBMS already obtains useful feedback while executing ordinary queries. For a filter predicate, the optimizer's estimated selectivity and the executor's observed selectivity expose a local discrepancy between the stale histogram and the current data distribution. OASIS turns a recent window of such observations into a corrected single-column histogram before the statistics reach the optimizer. The design is intentionally non-invasive: OASIS requires a query-completion feedback listener and a statistics-assembly hook, but no changes to the optimizer's search algorithm or cost model.

The central challenge is that feedback is sparse, local, and heterogeneous. Naively forcing a histogram to match a few recent predicates can overfit the feedback window; ignoring the feedback leaves the prior stale. OASIS addresses this problem by learning a *drift-correction function*. It encodes the stale quantile histogram together with recent predicate observations and uses a compact pooled MLP to predict corrected quantile boundaries. The same 38K-parameter checkpoint is trained once on simulation-derived drift scenarios and then reused across columns and schemas in our experiments.

Because OASIS operates at the marginal-statistics layer, its role in multi-column settings is conditional rather than automatic. It supplies corrected one-dimensional inputs; it does not infer dependence structure or repair join correlations by itself. The experiments show that learned global marginal repair and local feedback consistency are complementary. Plain OASIS improves direct single-column estimates under drift, but downstream estimators can amplify local CDF errors when the corrected marginal does not match the current feedback window. We therefore introduce OASIS-Proj, a lightweight calibration step that starts from the OASIS prediction and projects the corrected marginal onto current feedback constraints before the marginal is consumed by a downstream estimator or DBMS planner. In deployment, OASIS-Proj or residual-gated Hybrid is the recommended form; plain OASIS is the learned marginal-repair component.

The evaluation follows this scope. We evaluate the mechanism where it acts: future selectivity estimates, structural histogram error, downstream marginal composition, and real PostgreSQL planner estimates and plan shapes. The PostgreSQL study is planner-only: true rows are obtained with `COUNT(*)`, and optimizer-facing evidence comes from `EXPLAIN (FORMAT JSON)` under injected statistics states, without measuring query runtime. To reduce dependence on a single drift simulator, we also evaluate out-of-distribution drift families and benchmark-inspired DML traces with explicit insert, update, and delete events.

The contributions are:

- We formulate feedback-driven single-column statistics correction as a DBMS-facing problem: given a stale histogram and a causal window of predicate feedback, produce corrected statistics that improve future selectivity estimates without rescanning the table.
- We design OASIS, a lightweight learned correction model that encodes prior quantiles and heterogeneous observations into a fixed tensor and predicts corrected histogram boundaries with a pooled MLP.
- We instantiate a PostgreSQL-style statistics-format conversion interface that maps MCV+histogram statistics into a canonical quantile/CDF view and back, while isolating engine-specific adapter logic from the correction model.

- We characterize the single-column correction behavior of OASIS across drift intensity, structural metrics, initial-distribution transfer, unseen drift families, DML trace sanity checks, simple heuristics, and published feedback-driven baselines.
- We identify and address a deployment boundary for multi-column composition: learned marginals alone do not guarantee stable downstream gains when current feedback constraints are violated. OASIS-Proj combines learned global marginal repair with local feedback consistency before downstream use.
- We validate the DBMS optimizer path with a PostgreSQL planner-only statistics-injection experiment across 12 configurations, showing that corrected marginals move estimated rows and plan shapes toward the fresh-statistics planner without making runtime claims.
- We summarize deployment-safety evidence across PostgreSQL plan deviations, feedback-budget sensitivity, feedback-noise sensitivity, and trace-grounded drift, motivating projection and residual-based gating as the calibrated deployment form.

2. Background and Problem Formulation

2.1. Column Statistics and the Staleness Gap

We model a column histogram as an equi-depth quantile summary with B buckets. The internal boundaries $v_1, \dots, v_{B-1}$ correspond to fixed quantile levels $p_1, \dots, p_{B-1}$. A range predicate such as $x < c$ is estimated by evaluating the piecewise-linear CDF implied by those boundaries. Although real DBMS statistics may combine histograms with auxiliary summaries such as most-common-value lists, density vectors, or null fractions, OASIS operates on a canonical one-dimensional CDF view.

Let H_0 denote the stale histogram captured at the last statistics refresh and H^* denote the current post-drift distribution. For a predicate σ, the optimizer derives an estimate $\hat{s}_{H_0}(\sigma)$ while the executor can later reveal an actual selectivity $s^*(\sigma)$. We measure selectivity error using Q-error:

$$\mathrm{QErr}(\hat{s}, s^*) = \max\left(\frac{\hat{s}}{s^*}, \frac{s^*}{\hat{s}}\right). \tag{1}$$

2.2. Statistics Correction

Given a stale histogram H_0 and an observation window $\mathcal{O} = \{o_1, \dots, o_K\}$, where each observation records a predicate type, predicate boundary, estimated selectivity, and actual selectivity, the goal is to produce corrected statistics $\hat{H}$ such that future predicates are estimated more accurately:

$$\mathbb{E}_\sigma \left[\mathrm{QErr}\left(\mathrm{CDF}_{\hat{H}}(\sigma), s^*_\sigma\right)\right] < \mathbb{E}_\sigma \left[\mathrm{QErr}\left(\mathrm{CDF}_{H_0}(\sigma), s^*_\sigma\right)\right].$$

The expectation is over future predicates from the workload distribution.

The solution must satisfy three deployment constraints. First, it must avoid table rescans, deriving $\hat{H}$ only from H_0 and recent feedback. Second, it must be cheap enough to run near query-planning time. Third, it must degrade gracefully when feedback is sparse or uninformative, preserving the stale prior rather than applying large unsupported corrections.

2.3. Scope

OASIS targets single-column histogram staleness. It does not repair multi-column correlation errors, join-key dependence, physical design problems, or cost-model calibration. These terms remain in the optimizer's error budget even if the single-column histogram is accurate. The role of OASIS is to improve the marginal statistics that many downstream estimators and optimizers already consume.

3. OASIS Design

3.1. System Overview

OASIS sits between the statistics store and the optimizer. When the optimizer requests a histogram, OASIS reconstructs a canonical quantile representation, retrieves a recent observation window for the column, predicts corrected quantile boundaries, optionally applies a feedback-consistency projection, and returns an engine-native statistics object to the CBO. Query execution remains unchanged except for lightweight per-conjunct feedback collection.

3.2. Feature Representation

The correction model consumes a fixed tensor containing four blocks. The prior block stores the normalized quantile boundaries of H_0. The meta block stores the null fraction, observation count ratio, and bucket count. The observation block stores up to K recent predicate observations; each observation contains a predicate-type encoding and numeric features for predicate bounds, estimated selectivity, actual selectivity, and recency. The mask block marks valid observation slots. In our implementation $B = 10$ and $K = 16$, producing a compact input representation.

3.3. Correction Model

OASIS uses a pooled MLP with a prior encoder, an observation encoder, an attention-like pooling layer over the observation window, and a residual output head that predicts corrections to the internal quantile boundaries. Residual prediction encourages conservative behavior when the feedback window is uninformative. The predicted boundaries are clamped to the normalized domain and projected into monotone order to ensure a valid CDF.

The model is trained offline on simulation-derived compound drift. Each training sample contains a stale prior, a causal feedback window, and post-drift quantile targets. Once trained, the same checkpoint is reused across columns in the evaluation; no per-table or per-distribution retuning is used.

3.4. Feedback Collection and Integration

OASIS requires two integration points. A query-completion listener records predicate-level feedback from executor row-count instrumentation. A statistics-assembly hook intercepts the histogram before it reaches the optimizer and substitutes the corrected version. Because the correction is inserted at the statistics layer, the optimizer's search procedure and cost model remain unchanged.

3.5. Statistics-Format Conversion

Many DBMSs do not expose a standalone editable equi-depth histogram. PostgreSQL, for example, couples histogram boundaries with MCV lists and other per-column summaries. We instantiate the interface for PostgreSQL-style statistics and design it to isolate engine-specific adapters. The adapter has three phases: reconstruct a canonical distribution view from the native object, apply correction on the canonical view, and decompose the corrected view back into native statistics fields. This paper validates the PostgreSQL-style MCV+histogram instantiation; other engines require their own adapter logic.

3.6. OASIS-Proj: Feedback-Consistency Calibration

Plain OASIS predicts a global marginal correction from learned drift patterns. For planner use, however, the corrected CDF must also be reliable near the predicates that have just been observed. This local requirement is especially important before multi-column composition, where a small one-dimensional CDF mismatch can become a larger joint-selectivity error.

OASIS-Proj therefore applies a lightweight projection after the OASIS prediction. The projection uses the OASIS histogram as the starting point and adjusts it to satisfy the current feedback constraints as closely as possible under an ISOMER/IPF-style consistency principle. This step remains a single-column calibration layer; it does not estimate multi-column dependence.

4. Experimental Methodology

The evaluation asks five questions. First, does the statistics-format conversion preserve probability mass and selectivity estimates? Second, how much does OASIS improve single-column histogram quality under drift? Third, when do corrected marginals compose safely with downstream multi-column estimators, and when is feedback consistency required? Fourth, do corrected statistics affect a real PostgreSQL optimizer pipeline in the intended direction, without measuring runtime? Fifth, what safety checks are needed when feedback is sparse, noisy, or trace-like rather than drawn from the training generator?

Unless otherwise stated, OASIS is trained once on Gaussian compound drift with $q \in \{1, 3, 5, 10, 15, 20\}$ and evaluated on held-out drift intensities $q \in \{1, 3, 5, 10, 15, 20, 25, 30\}$. All feedback-driven methods consume the same observation budget $K = 16$ and emit corrected quantiles on the same $B = 10$ grid. We compare against the stale prior, STHoles [1], ISOMER [2], QuickSel-H [3], and simple feedback heuristics. Metrics include Q-error, selectivity MAE, quantile MAE, feedback residual, PostgreSQL estimated-row Q-error, and plan-shape match to the fresh-statistics planner. For deployment-sensitivity diagnostics that require many perturbed runs, we also use a transparent optimizer-decision proxy: future predicates are generated from held-out distributions, each statistics state selects scan and join choices under a fixed cost proxy, and the selected choice is scored by its true proxy cost. This proxy is used only to analyze feedback-budget and feedback-noise behavior; it is not a runtime benchmark.

5. Single-Column Statistics Correction

5.1. Format-Conversion Validation

The PostgreSQL-style MCV+histogram conversion preserves total probability mass exactly in the tested settings. Round-trip residual-quantile MAE

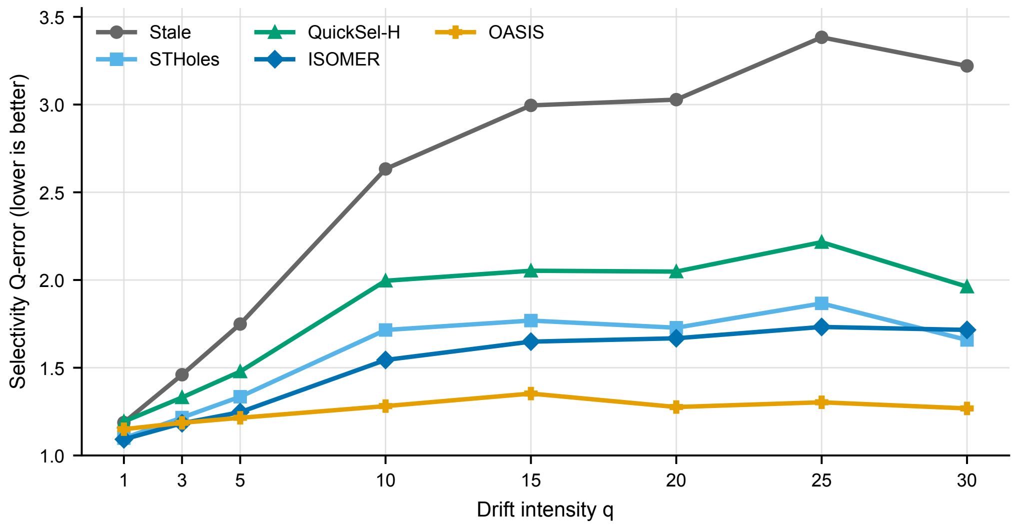

Figure 1: Single-column selectivity Q-error across drift intensity. All methods consume the same feedback budget. OASIS remains close to the stale prior at mild drift and becomes strongest as drift grows.

is at machine precision, with mean 2.4×10^{-17} and maximum 9.2×10^{-17}. Selectivity estimates remain consistent before and after conversion, with global MAE 1.1×10^{-4} and worst-case error below 0.39%. With PostgreSQL-like defaults, the full round trip completes in 0.066–0.073 ms. This validates the PostgreSQL-style conversion as a low-overhead bridge between native statistics and the canonical OASIS view.

5.2. Q-Error Across Drift Intensity

Figure 1 reports the main single-column result. At mild drift ($q = 1$), ISOMER is strongest and OASIS remains close to the stale prior, consistent with the graceful-degradation objective. From $q \geq 5$, OASIS is the best method under the shared feedback budget. At $q = 10$, 20, and 30, OASIS reduces Q-error over the stale prior by 51.3%, 57.9%, and 60.6%, respectively. The advantage persists on unseen drift intensities $q = 25$ and $q = 30$.

Table 1 compares OASIS with simple feedback heuristics. Fixed linear interpolation and exponential feedback averaging are substantially weaker than learned correction; FeedAvg worsens estimation in all tested settings. This shows that the gains are not simply due to using feedback, but to learning how the prior and observations should interact.

Table 1: Simple heuristic baselines vs. OASIS at selected drift intensities. Results are averaged over three independent training seeds (42, 123, 456) with a different evaluation protocol than the main comparison in Table **??**; absolute Q-Error values are therefore not directly comparable across the two tables. Both heuristics consume the same observation window K=16. FeedAvg worsens estimation at all drift levels.

	Q-Error (↓ better)			
q	Stale	LinInterp	FeedAvg	OASIS
5	1.791	1.628	1.835	**1.238**
10	2.471	2.118	2.565	**1.255**
20	3.370	2.749	3.553	**1.296**
30	3.500	2.823	3.690	**1.345**

5.3. Structural Metrics and Initial-Distribution Generalization

Table 2: Quantile MAE and Selectivity MAE across drift intensities from the clean synthetic rerun. Bold denotes best method per metric.

	Quantile MAE (↓)					Selectivity MAE (↓)				
q	Prior	STHoles	QSel-H	ISOMER	OASIS	Prior	STHoles	QSel-H	ISOMER	OASIS
1	0.043	0.027	0.036	**0.023**	0.041	0.038	0.024	0.031	**0.020**	0.035
3	0.106	0.064	0.071	**0.047**	0.048	0.095	0.057	0.064	**0.040**	0.042
5	0.145	0.085	0.094	0.057	**0.054**	0.131	0.077	0.085	0.051	**0.048**
10	0.208	0.115	0.126	0.080	**0.055**	0.190	0.105	0.114	0.072	**0.049**
15	0.230	0.119	0.128	0.094	**0.062**	0.203	0.104	0.113	0.083	**0.054**
20	0.237	0.115	0.127	0.102	**0.052**	0.214	0.104	0.115	0.091	**0.046**
25	0.248	0.116	0.131	0.095	**0.053**	0.223	0.104	0.118	0.084	**0.046**
30	0.231	0.103	0.111	0.103	**0.049**	0.204	0.091	0.098	0.090	**0.042**

Table 2 shows that OASIS improves structural histogram quality, not only Q-error. At $q = 10$, quantile MAE drops from 0.208 for the stale prior to 0.055 for OASIS, and selectivity MAE drops from 0.190 to 0.049. OASIS remains the best method on both metrics for every $q \geq 5$.

Table 3 evaluates transfer across initial distributions at $q = 10$. A model trained on Gaussian-mixture initial states remains best across uniform, skewed, bimodal, triangular, exponential, and Gaussian-mixture starts, reducing Q-error by 44%–61% over the stale prior.

Table 3: Q-Error across different initial distributions at q=10. A model trained only on Gaussian-mixture initial states remains best on all tested distributions.

Initial Distribution	Stale	STHoles	QuickSel-H	ISOMER	OASIS
Gaussian Mixture	2.497	1.453	1.845	1.493	**1.262**
Uniform	2.409	1.503	1.977	1.534	**1.284**
Skewed (Power-law)	3.517	1.679	2.202	1.609	**1.379**
Bimodal	2.488	1.554	2.069	1.462	**1.276**
Triangular	2.225	1.420	1.833	1.505	**1.238**
Exponential	3.097	1.592	2.141	1.593	**1.292**

5.4. Out-of-Distribution Drift Realism

The main OASIS checkpoint is trained on compound drift. To test whether the learned correction is useful beyond that generator, we evaluate it unchanged on six deployment-inspired drift families: batch loads, range shifts, skew evolution, outlier bursts, multimodal drift, and seasonal/mixed drift. Each case starts from a stale histogram, interleaves the new drift family with 16 feedback observations, and evaluates future selectivity against the final post-drift distribution. No OOD retraining is used.

Figure 2 shows that the correction path remains useful under these unseen drift mechanisms. The strongest OOD stress case is batch loading, where stale Q-error is 4.203; plain OASIS reduces it to 1.280, outperforming ISOMER's 1.461, and OASIS-Proj further improves to 1.272. On range shift, skew evolution, outlier, multimodal, and seasonal drift, OASIS-Proj is the best non-fresh method, with Q-error between 1.057 and 1.187. This experiment is not a substitute for production trace validation, but it shows that the gains are not confined to the compound-drift generator used for training.

5.5. Benchmark-Inspired DML Trace Sanity Check

The OOD suite changes the drift family, but it still uses synthetic drift operators. We therefore add a second realism check with explicit DML event streams anchored to analytical benchmark table/column patterns: append-heavy sales dates, promotion price revisions, inventory restocking, returns and cancellations, customer segment churn, and seasonal mixed maintenance. Each trace emits insert, update, and delete counts before a feedback observation is collected. These traces are not production logs or DBMS telemetry;

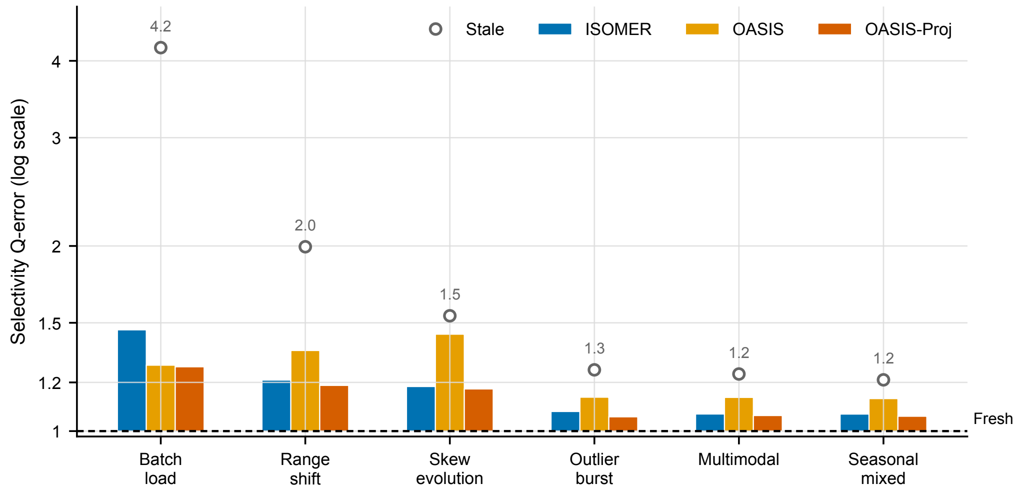

Figure 2: Out-of-distribution drift realism suite. OASIS is trained only on compound drift and evaluated unchanged on deployment-inspired drift families. Values are geometric mean selectivity Q-error on a log-scaled axis; stale statistics are shown as open circles and the dashed line is fresh post-drift statistics.

they are a transparent sanity check that the correction layer remains useful when drift is produced by event sequences rather than by the compound training generator.

Figure 3 reinforces the same deployment pattern under event-sequence drift. Across the six traces, stale statistics have aggregate Q-error 1.295; OASIS-Proj reduces this to 1.076 and residual-gated Hybrid to 1.072. OASIS-Proj is the best non-fresh method on sales append and returns/cancellation traces, where the stale histogram is visibly shifted. On the milder price, inventory, customer, and seasonal traces, ISOMER is best or tied because enforcing the current feedback window is sufficient. Plain OASIS can regress on these mild update-heavy traces, which supports the deployment policy: learned global repair is useful, and the statistics object passed to a planner should be projected or residual-gated.

6. Optimizer-Facing Evaluation

6.1. Counterfactual Selectivity Evaluation

We first isolate the causal effect of the statistics state on selectivity estimation. The same held-out drift samples are evaluated under stale, OASIS-

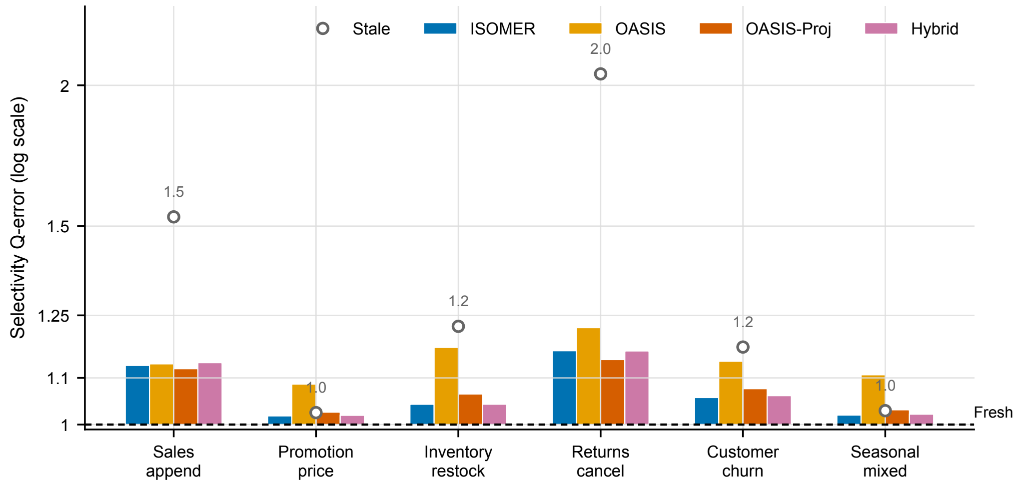

Figure 3: Benchmark-inspired DML trace sanity check. Each trace emits explicit insert/update/delete events anchored to an analytical benchmark maintenance pattern. Values are geometric mean selectivity Q-error on a log-scaled axis; stale statistics are shown as open circles and the dashed line is fresh post-drift statistics.

corrected, and simulated sampled-refresh statistics, while the underlying post-drift distribution and predicates are held fixed. This is not a runtime benchmark; it measures selectivity Q-error under controlled statistics states.

Table 4 shows that OASIS recovers 64%–77% of the fresh/oracle improvement and substantially outperforms the sampled-refresh proxy. The result should be interpreted as per-predicate selectivity evidence, not as evidence of wall-clock query acceleration.

6.2. Marginal-to-Joint Composition

This experiment studies composition rather than replacing multi-column cardinality-estimation methods. OASIS repairs one-dimensional marginals, so the question is how those repaired marginals behave when a downstream estimator combines them under a fixed dependence assumption. We generate correlated two-column tables with drift, correct each single-column marginal independently, and feed the marginals into two joint estimators: independence, the maximum-entropy estimate under only one-dimensional marginal constraints, and a Gaussian copula with fixed correlation structure. Fresh uses post-drift marginal inputs under the same downstream estimator; it is not an oracle joint distribution.

Table 4: Counterfactual selectivity evaluation on compound drift test data. "Sampled-Refresh" is a simulated column-level resampling proxy. Recovery measures the fraction of fresh/oracle Q-error improvement recovered by OASIS. No query runtime is measured.

q	#Pred	Stale QE	Sampled-Refresh	OASIS QE	Recovery
5	30	1.714	1.492	**1.263**	69%
10	30	2.720	2.121	**1.232**	77%
15	30	2.798	2.183	**1.284**	74%
20	30	2.779	2.201	**1.389**	64%
Mean	120	2.503	1.999	**1.292**	71%

Table 5: Marginal-to-joint composition: average Q-error (↓) for two-column range predicates. Values average configuration-level geometric means over 20 correlation/drift settings. OASIS-Proj is a feedback-consistency calibration layer on top of OASIS marginals; Hybrid uses only feedback-window residuals for selection.

Estimator	Stale	OASIS	ISOMER	OASIS-Proj	Hybrid	Fresh	OASIS +%	Proj +%	Hybrid +%
Independence / max-entropy	1.820	1.679	1.220	1.222	**1.214**	1.185	+7.9%	+31.8%	+32.2%
Gaussian copula	1.558	1.513	1.185	1.176	**1.166**	1.139	+2.4%	+23.1%	+23.8%

Table 5 gives the main diagnostic result. Despite strong single-column performance, plain OASIS provides only modest and unstable gains after two-column composition: it improves average joint Q-error by 7.9% under independence and by 2.4% under the Gaussian copula. ISOMER is substantially stronger in this setting because it enforces current feedback constraints. The attribution explains the gap: plain OASIS improves marginal Q-error over stale in only 43.6% of composition rows and beats ISOMER on marginal Q-error in only 3.9%. Thus, the learned marginal-repair component should not be interpreted as a stable multi-column correction method on its own.

The system conclusion is constructive: learned global marginal repair should be calibrated before composition. OASIS-Proj addresses this failure mode without changing the dependence model. By projecting OASIS-corrected marginals onto the feedback constraints, it reaches Q-error 1.222 under independence and 1.176 under the Gaussian copula, nearly matching ISOMER and approaching fresh marginal inputs. Hybrid provides a conservative deployment policy by choosing among stale, ISOMER, OASIS, and OASIS-Proj using only feedback-window residuals.

Table 6: Multi-configuration PostgreSQL planner-only statistics-injection evidence. Values are means over configurations; parentheses show one standard deviation across configurations. True rows use `COUNT(*)`; estimated rows and plan shapes use `EXPLAIN (FORMAT JSON)`. No query runtime is measured.

Method	Row QE	Fresh Plan	Recovery	New Deviations
Stale	28.436 (6.585)	56.0% (5.5)	0.0% (0.0)	0.0% (0.0)
ISOMER	2.372 (0.283)	96.1% (1.9)	92.2% (3.9)	1.0% (1.0)
OASIS	3.313 (0.228)	88.6% (2.7)	91.4% (3.2)	13.7% (2.0)
OASIS-Proj	2.375 (0.275)	96.1% (1.9)	92.2% (3.9)	1.0% (1.0)
Hybrid	2.374 (0.286)	96.1% (1.9)	92.2% (3.9)	1.0% (1.0)
Fresh	1.415 (0.095)	100.0% (0.0)	100.0% (0.0)	0.0% (0.0)

6.3. PostgreSQL Planner-Only Statistics Injection

The previous experiments evaluate selectivity and downstream estimator behavior. We next validate that corrected statistics can affect a real DBMS optimizer pipeline. The PostgreSQL experiment creates a synthetic table with fresh current data, builds alternative single-column statistics for `fact.x`, and injects them into `pg_statistic` by copying typed statistics rows from analyzed source tables. For each query, true cardinality is obtained with `COUNT(*)`; PostgreSQL estimated rows and plan shape are obtained with `EXPLAIN (FORMAT JSON)`. The experiment does not measure query runtime.

Table 6 provides the strongest DBMS-facing evidence in this paper. The batch covers two drift directions, two table sizes, and three random seeds, yielding 12 PostgreSQL configurations. Across all query instances, stale statistics produce aggregate row Q-error 27.748 and match the fresh-statistics plan shape on only 56.0% of queries. Plain OASIS reduces row Q-error to 3.306 and recovers 91.7% of stale/fresh plan-shape disagreements, demonstrating that learned marginal correction propagates into PostgreSQL's native planner. However, it also introduces new plan deviations in 13.8% of cases where stale statistics already matched fresh.

OASIS-Proj largely removes this safety issue. It reduces aggregate row Q-error to 2.360, matches the fresh-statistics plan shape on 96.1% of queries, recovers 92.6% of stale/fresh plan-shape disagreements, and reduces new plan deviations to 1.1%. Across configurations, mean row Q-error is 2.375 ± 0.275 for OASIS-Proj, 2.372 ± 0.283 for ISOMER, and 3.313 ± 0.228 for plain OASIS. The planner-only result supports the deployment story: OASIS provides

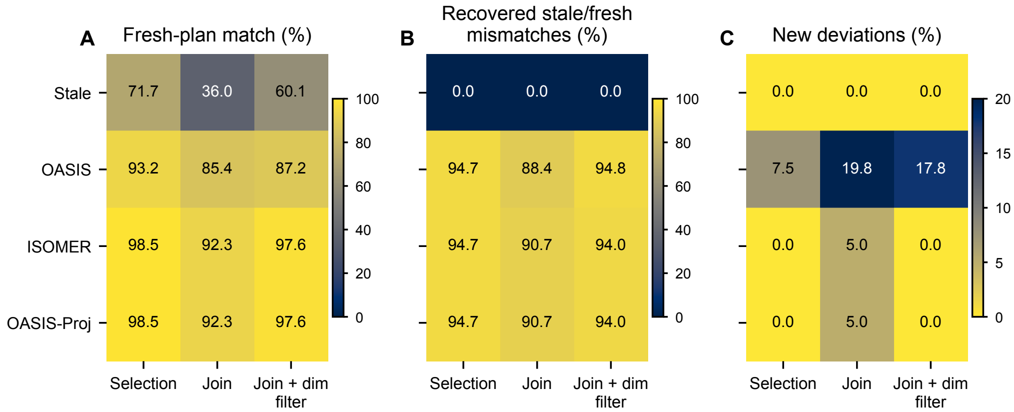

Figure 4: PostgreSQL planner-only plan-shape effects by query family. Cell labels are percentages. Panel A reports fresh-statistics plan-shape match, Panel B reports recovery of stale/fresh mismatches, and Panel C reports new deviations where stale already matched fresh.

learned global marginal correction, while OASIS-Proj calibrates the corrected marginal before the DBMS planner consumes it.

Figure 4 breaks the PostgreSQL result down by query family. Stale statistics disagree with the fresh-statistics plan on 95/336 selection queries, 215/336 join queries, and 134/336 join-with-dimension-filter queries. OASIS-Proj matches the fresh plan on 98.5% of selections, 92.3% of joins, and 97.6% of filtered joins. The safety benefit over plain OASIS is clearest in new deviations: plain OASIS creates new plan-shape deviations in 7.5%, 19.8%, and 17.8% of the three families, whereas OASIS-Proj reduces these rates to 0.0%, 5.0%, and 0.0%.

6.4. Feedback Budget and Noise Sensitivity

The deployment form of OASIS depends on two practical conditions: how many recent feedback observations are available and how accurate those feedback observations are. To stress these conditions without running additional DBMS loads, we use the optimizer-decision proxy described in Section 4. All methods see the same feedback window and future predicates; only the feedback budget or feedback values are changed.

Panels A–B of Figure 5 vary the number of visible feedback observations. With only two observations, consistency projection is helpful but limited: OASIS-Proj reaches selectivity Q-error 2.233 and Hybrid matches 84.6%

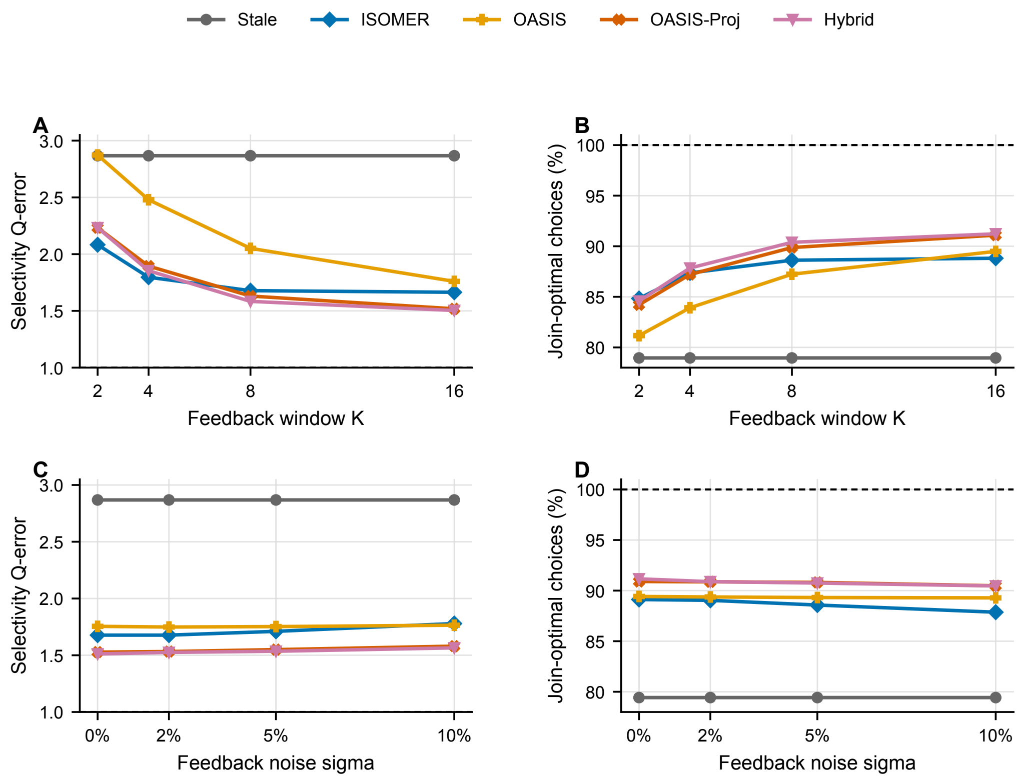

Figure 5: Feedback-budget and feedback-noise sensitivity in the optimizer-decision proxy. Panels A–B vary the visible feedback window K; Panels C–D perturb feedback selectivities with multiplicative log-normal noise while holding future predicates fixed. Dashed lines denote fresh statistics.

of optimal join choices, compared with stale Q-error 2.867 and 79.0% join-optimal choices. As the window grows, the learned correction and projection become more complementary. At $K = 16$, OASIS-Proj reaches Q-error 1.518 and Hybrid reaches 1.504, with 91.1%–91.2% join-optimal choices. Hybrid's selections also show that no single component dominates all cases: at $K = 16$, it selects ISOMER for 37% of cases, OASIS-Proj for 47%, plain OASIS for 14%, and stale for 2%.

Panels C–D perturb feedback selectivities with multiplicative log-normal noise while keeping the future predicates and true selectivities fixed. The diagnostic uses 64 held-out cases per drift intensity and one noise seed. At 10% feedback noise, OASIS-Proj still improves selectivity Q-error from 2.868

Table 7: Deployment safety checks. Projection and residual-based Hybrid gating are evaluated without oracle/fresh information. P/I/O/S denote OASIS-Proj, ISOMER, OASIS, and stale choices in the residual gate.

Check	Risk signal	Unguarded state	Calibrated/gated state	Gate signal
PostgreSQL planner	New plan deviations	OASIS 13.8%	OASIS-Proj 1.1%; FreshPlan 96.1%	projection
PostgreSQL joins	Join new deviations	OASIS 19.8%	OASIS-Proj 5.0%; Recovery 90.7%	projection
Sparse feedback ($K = 2$)	Join-optimal choices	Stale 79.0%	Hybrid 84.6%; NewRisk 2.5%	I 27%, P 57%, O 14%, S 1%
Full feedback ($K = 16$)	Join-optimal choices	Stale 79.0%	Hybrid 91.2%; NewRisk 1.4%	I 37%, P 47%, O 14%, S 2%
10% feedback noise	Join-optimal choices	Stale 79.4%	Hybrid 90.5%; NewRisk 1.7%	I 36%, P 42%, O 18%, S 3%
DML trace sanity	Selectivity Q-error	Stale 1.295	Hybrid 1.072; Proj 1.076	I 77%, P 20%, O 2%, S 1%

to 1.580 and maintains 90.5% join-optimal choices; Hybrid reaches Q-error 1.565 with the same join-optimal rate. ISOMER is more sensitive to noise in this setting, rising from Q-error 1.677 at clean feedback to 1.779 at 10% noise. The result supports the role of OASIS-Proj as a calibration layer, but it also motivates residual-based gating and abstention when feedback attribution is unreliable.

6.5. Deployment Safety Summary

Table 7 collects the safety evidence that determines how OASIS should be deployed. The unguarded learned correction improves row-estimation accuracy, but it can introduce plan deviations when the stale planner already agreed with fresh statistics. Projection reduces these new deviations from 13.8% to 1.1% in the PostgreSQL batch and from 19.8% to 5.0% on join queries. In the optimizer-decision proxy, residual-gated Hybrid improves join-optimal choices under sparse feedback, full feedback, and 10% feedback noise while keeping new risky decisions near 1–3%. On the benchmark-inspired DML traces, Hybrid improves aggregate selectivity Q-error from 1.295 to 1.072 and selects ISOMER or OASIS-Proj in 97% of cases. These results make OASIS-Proj and residual gating part of the deployment design, rather than optional post-processing.

7. Overhead and Deployment Considerations

Per-correction OASIS inference averages 1.06 ms in our implementation, including tensorization and one forward pass. The PostgreSQL-style statistics-format conversion adds 0.066–0.073 ms, for a combined correction cost of approximately 1.13 ms. Feedback collection adds bounded bookkeeping per filter conjunct and emits one observation tuple per conjunct at query completion. These measurements support the intended use case: maintaining optimizer-facing statistics between scheduled statistics refreshes, not replacing full refreshes after major schema changes or bulk data shifts.

8. Related Work

Classical histogram construction builds statistics from data scans [4, 5], while streaming summaries such as KLL sketches provide compact approximate quantiles [6]. Adaptive selectivity estimation has long used query feedback to refine estimation functions [7]. Feedback-driven histograms, including self-tuning histograms, STHoles, and ISOMER, refine statistics from observed query results [8, 1, 2]. OASIS differs by learning a reusable drift-correction function that emits DBMS-consumable statistics rather than applying only local per-table refinement rules or returning point selectivity estimates.

Learned cardinality estimators such as NeuroCard, Naru, DeepDB, FACE, and MSCN predict query cardinalities using learned density or representation models [9, 10, 11, 12, 13]. These systems are complementary to OASIS: they often output point estimates for specific query classes, while OASIS outputs corrected marginal statistics that can be consumed by an existing CBO. Copula-based methods similarly combine one-dimensional marginals with dependence models; our composition experiments show that marginal quality and feedback consistency are prerequisites for such downstream use.

Plan-level optimization systems such as LEO, Neo, and Bao adjust plan choices or optimizer behavior based on execution history [14, 15, 16]. OASIS operates below that layer by correcting the statistics object itself. A system could combine both approaches: statistics correction supplies better base-table inputs, while plan-level learning handles residual cost-model or join-order effects.

9. Limitations

The evaluation is limited to tested synthetic drift families, benchmark-inspired DML traces, and controlled PostgreSQL planner-only scenarios. The DML traces make insert/update/delete event streams explicit, but they are not production logs or DBMS telemetry; validating the same correction policy on production statistics histories and workload traces remains future work. The PostgreSQL batch covers two drift directions, two table sizes, and three random seeds, so it validates the statistics-to-planner path more robustly than a single case, but it does not establish wall-clock runtime improvement. Runtime depends on executor behavior, caching, physical design, system load, and cost-model calibration. OASIS also does not infer multi-column dependence from single-column feedback. When downstream estimates depend on correlations, an external dependence model or multivariate statistics source is still required. Finally, feedback itself can be noisy because of predicate attribution, sampling, or concurrent updates. Our noise diagnostic shows graceful degradation up to 10% multiplicative feedback noise, but deployments should pair correction with residual-based gating, abstention, and rollback policies.

10. Conclusion

OASIS repairs stale single-column optimizer statistics from query feedback without rescanning tables or modifying the optimizer. A compact learned model predicts corrected quantile boundaries from a stale prior and recent observations, while a PostgreSQL-style statistics adapter connects the canonical correction view to DBMS-native statistics. Across synthetic drift scenarios, OOD drift families, and benchmark-inspired DML traces, the corrected statistics improve single-column selectivity estimates and histogram structure.

The broader lesson is that learned marginal correction is valuable when it is deployed at the right layer. OASIS is not a replacement for multi-column cardinality estimation, and it does not solve correlation errors on its own. Instead, it supplies corrected marginal statistics that downstream estimators and planners can consume. OASIS-Proj adds the feedback-consistency layer needed for that use: OASIS supplies learned global marginal repair, while the projection enforces the local constraints that downstream estimators and planners are likely to trust. Accordingly, the recommended deployment path is OASIS-Proj when a calibrated corrected histogram is needed, or residual-gated Hybrid when the system can choose among stale, ISOMER, OASIS, and OASIS-Proj from the current feedback residual. In the 12-configuration PostgreSQL planner-only batch, OASIS-Proj reduces aggregate row-estimation Q-error from 27.748 to 2.360, matches the fresh-statistics plan shape on 96.1% of queries, and keeps new plan deviations near 1%, without measuring or claiming query runtime improvement. In the trace-grounded sanity check, residual-gated Hybrid reduces aggregate selectivity Q-error from 1.295 to 1.072. This makes OASIS a practical component for maintaining optimizer-facing statistics between scheduled `ANALYZE` passes.

References

[1] N. Bruno, S. Chaudhuri, L. Gravano, STHoles: A multidimensional workload-aware histogram, in: Proceedings of SIGMOD, 2001, pp. 211–222.

[2] V. Markl, N. Megiddo, M. Kutsch, T. M. Tran, P. Lang, V. Raman, Consistent selectivity estimation via maximum entropy, The VLDB Journal 16 (2) (2007) 169–188.

[3] Y. Park, S. Zhong, B. Mozafari, QuickSel: Quick selectivity learning with mixture models, in: Proceedings of SIGMOD, 2020, pp. 1017–1033.

[4] Y. E. Ioannidis, V. Poosala, Universality of serial histograms, in: Proceedings of the 19th VLDB Conference, 1993, pp. 256–267.

[5] V. Poosala, Y. E. Ioannidis, Selectivity estimation without the attribute value independence assumption, in: Proceedings of the 23rd VLDB Conference, 1997, pp. 486–495.

[6] N. Ivkin, J. Nelson, H. Yu, V. Braverman, KLL: Approximately optimal streaming quantile estimation, Proceedings of the ACM on Management of Data 2 (1) (2019) 1–23.

[7] C. M. Chen, N. Roussopoulos, Adaptive selectivity estimation using query feedback, in: Proceedings of the 24th ACM SIGMOD International Conference on Management of Data, ACM Press, 1994, pp. 161–172. `doi:10.1145/191839.191874`.

[8] A. Aboulnaga, S. Chaudhuri, Self-tuning histograms: Building histograms without looking at data, in: ACM SIGMOD Record, Vol. 28, 1999, pp. 181–192.

[9] Z. Yang, A. Kamsetty, S. Luan, E. Liang, Y. Duan, X. Chen, I. Stoica, NeuroCard: One cardinality estimator for all tables, in: Proceedings of the VLDB Endowment, Vol. 14, 2020, pp. 61–73.

[10] Z. Yang, E. Liang, A. Kamsetty, C. Wu, Y. Duan, X. Chen, P. Abbeel, J. M. Hellerstein, S. Krishnan, I. Stoica, Deep unsupervised cardinality estimation, Proceedings of the VLDB Endowment 13 (3) (2019) 279–292.

[11] B. Hilprecht, A. Schmidt, M. Kulessa, A. Molina, K. Kersting, C. Binnig, DeepDB: Learn from data, not from queries!, in: Proceedings of the VLDB Endowment, Vol. 13, 2020, pp. 992–1005.

[12] Z. Yang, E. Liang, Y. Duan, S. Luan, I. Stoica, J. M. Hellerstein, FACE: A normalizing flow based cardinality estimator, in: SIGMOD Workshop on Approximate Query Processing, 2020.

[13] A. Kipf, T. Kipf, B. Radke, V. Leis, P. Boncz, A. Kemper, Learned cardinalities: Estimating correlated joins with deep learning, in: CIDR, 2019.

[14] M. Stillger, G. Lohman, V. Markl, M. Kandil, LEO – DB2's learning optimizer, in: Proceedings of the 27th VLDB Conference, 2001, pp. 19–28.

[15] R. Marcus, P. Negi, H. Mao, C. Zhang, M. Alizadeh, T. Kraska, O. Papaemmanouil, N. Tatbul, Neo: A learned query optimizer, in: Proceedings of the VLDB Endowment, Vol. 12, 2019, pp. 1705–1718.

[16] R. Marcus, P. Negi, H. Mao, N. Tatbul, M. Alizadeh, T. Kraska, Bao: Making learned query optimization practical, in: Proceedings of SIGMOD, 2022, pp. 1275–1288.